

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 19

Team Members :

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1. Problem Title : Smart Agriculture Monitoring System

2. Problem Description: Traditional farming lacks timely data decision making

3. Identified Entities (Classes)

Class Name	Description
Crop	Stores crop conditions (water, insects, pesticide). Calculates health based on internal state and soil fertility. Drives system output through its health value.
Soil	Represents soil fertility. Supplies fertility data used in crop health calculation.
Farmer	Manages crop and soil data and land. Provides data for analysis and applies received advice. The main interaction point in the system.
System	Processes farmer data to calculate yield. Uses crop health as the primary factor. Contains only computation logic.
Expert	Analyzes yield and crop conditions. Generates structured advice for improvement. Sends advice to Farmer and Admin.
Advice	Contains a String message for readable guidance. Represents Expert recommendations in structured form.
Admin	Stores all generated advice as logs. Maintains basic system records.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Crop	name waterGiven pesticideGiven insects waterReq health	deployhelp() changeConditionshelp() calculateHealth()
Soil	name soilFertility	fertilize()
Farmer	farmerCounter farmerId fertilizer yield soil crop recommendation	calculateYield() recieveAdvice() actOnAdvice() deploy() ChangeCondiitonshelp()
System	expert admin	analyzeConditions()
Expert	advice	generateAdvice()
Advice	waterChange pesticideChange fertilizerChange recommendation	generateRecommendation() dispense()
Admin	recordOfAdvice	storeAdvice() displayRecommendations()